



Identification of a murder caused by brodifacoum poisoning based on clinical examinations and LC-MS/MS results

Zhonghao Yu¹ · Erjuan Zhao² · Qing Shi¹ · Yuhao Yuan¹ · Jinghong Ma² · Luwang Zhou³ · Yijie Duan¹ · Yiwu Zhou¹

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Abstract

Brodifacoum exerts its antagonistic effect against the metabolism of vitamin K, an essential component in the synthesis of blood coagulation factors. This effect ultimately hinders the blood's capacity to clot effectively, rendering it a commonly employed rodenticide. Instances of lethal poisonings are exceedingly rare owing to expeditious medical intervention and treatment. Within this report, we present a case of brodifacoum-induced homicide, wherein the patient exhibited distinct clinical examinations and symptoms. Moreover, the patient's blood sample exhibited a noteworthy brodifacoum concentration of 0.681 µg/mL even after a period of 43 days following the incident of poisoning. Although an autopsy was not conducted due to religious restrictions, we endeavor to reasonably deduce the cause of death and furnish corroborative evidence for clinical diagnosis, treatment, and forensic examination in instances involving brodifacoum poisoning.

Keywords Brodifacoum poisoning · Anticoagulant · Death · LC-MS/MS

Introduction

Brodifacoum, an extended-acting rodenticide with anticoagulant properties, is widely employed for the eradication of rats. Research has demonstrated that, in typical physiological circumstances, vitamin K is essential to the blood clotting cascade by serving as a cofactor for the γ -carboxylation of vitamin K-dependent coagulation factors (including factors VII, IX, and X) generated in the liver [1]. However, brodifacoum disrupts this process by irreversibly binding with the vitamin K epoxide reductase complex, inhibiting the conversion of vitamin K from epoxide to hydroquinone. Consequently, this disrupts the formation of vitamin K-dependent coagulation factors (VII, IX and X), protein C, and protein S, leading to an abnormal blood clotting function [2].

The majority of cases involving brodifacoum poisoning are accidental in childhood exposures and intentional in adults, mainly through ingestion, inhalation or transcutaneous exposure [3, 4]. However, deaths caused by brodifacoum poisoning are still rare due to the timely and long-term effective treatments. In this paper, a murder by brodifacoum poisoning is presented through a comprehensive analysis of the medical records, the quantitative results of brodifacoum detection by LC-MS/MS method in the victim's blood sample prior to death, and the case investigation, even the

✉ Yijie Duan
gerichtsarzt@hotmail.com

✉ Yiwu Zhou
zhouyiwu@hust.edu.cn

Zhonghao Yu
1259970467@qq.com

Erjuan Zhao
2743136264@qq.com

Qing Shi
806496165@qq.com

Yuhao Yuan
362736952@qq.com

Jinghong Ma
78311559@qq.com

Luwang Zhou
345855598@qq.com

¹ Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, China

² Hubei Chongxin Forensic Center, Wuhan 430415, China

³ Chaisang District Public Security Bureau of Jiujiang City, 332100 Jiujiang, China

autopsy was not performed due to religious reasons. Considering that there are a few reports of murder by brodifacoum poisoning and the insufficiency of brodifacoum detection by LC-MS/MS method in the available literatures, we aim to provide these data and hope they can assist in the clinical diagnosis, treatment, and forensic analysis of similar brodifacoum poisoning cases.

Case report

Case history A 55-year-old man was intentionally poisoned by a co-worker using a bottle of coke laced with brodifacoum. Approximately 10 days after consuming the brodifacoum-laced coke, the man noticed blood in his urine. Thirteen days after the poisoning, he sought medical assistance and was admitted to the hospital. However, the man chose to leave the hospital after two weeks of treatment, believing that he was improving and due to financial constraints.

Forty-three days after the poisoning, the man was readmitted to the hospital due to the presence of extensive ecchymosis on his left arm and left hip, which had persisted for a week without resolving. However, after receiving appropriate treatment, he was discharged on the same day.

Sixty days after the poisoning, the man suddenly experienced a loss of consciousness and subsequently passed away. Due to religious reasons, the family declined an autopsy.

Figure 1 summarizes the timeline of events from the man's consumption of the brodifacoum-laced coke to his

death, including the pathological symptoms observed, relevant clinical examinations and treatments received.

Clinical examination Between the 13th and 43rd days after the man drank the coke containing brodifacoum, some multidimensional examinations of his physical condition were conducted, mainly including hemagglutination, blood routine and biochemical and urine routine tests, which were summarized in chronological order as shown in Table 1.

In order to show the trends of the above parameters more clearly, a visual analysis of some representative indicators is performed as Fig. 2. It is clear from the Fig. 2 that some raised indicators were gradually decreasing, except for the Neu%, indicating that the condition was improving with the treatments in the hospital (from the 13d to the 22d after drinking the coke).

Toxicological analyses

The blood collected for routine and biochemical test on the 43rd day after the poisoning was tested for brodifacoum by LC-MS/MS by the following procedures:

Sample preprocessing 1mL of blood was collected and added with 10 μ L of internal standard solution (warfarin-d5, 1 μ g/mL). Then, 3 mL of ethyl acetate (SCRC, China) was added into the mixture and vortexed. After centrifugation at 700 $\times$ g (DLAB, China) for 3 min, the supernatant was transferred to another centrifuge tube (EPPENDORF, Germany). Another 3 mL of ethyl acetate (SCRC, China) was added and the extraction was repeated. The organic phases

Days after drinking the coke						
10d 13d 27d 36d 38d 43d 60d						
Symptoms	Blood in urine	\	Ecchymosis on his left arm and left hip			Died
Examinations	\	Blood routine and biochemical tests	\	\	Blood routine and bio-chemical tests	\
	\	Hemagglutination test	\	Hemagglutination test	Hemagglutination test	\
	\	Urine routine test	\	\		\
Primary treatments (by i.v.gtt)	\	Fresh frozen plasma without white cell	\	\	Fresh frozen plasma without white cell	\
	\	Vitamin K ₁		\	Vitamin K ₁	\

Fig. 1 Some pathological symptoms, examinations, and treatments the man appeared or received from the day he drank the coke containing brodifacoum to death. (Note: the “\” represents that there is no record.)

(Abbreviations: *i.v.gtt*: intravenously guttae, as the same meaning as intravenous drip.)

Table 1 Some significant results of clinical examinations

Days after drinking the coke	13d	13.5d	14d	14.5d	16d	22d	25d	26d	38d	43d
APTT (21–37 S)	110.37 ↑	87.38 ↑	40.33 ↑	40.94 ↑	42.94 ↑	41.83 ↑	56.28 ↑	43.62 ↑	165.21 ↑	>200 ↑
PT (10–14 S)	>100 ↑	>100 ↑	14.63 ↑	19.16 ↑	20.96 ↑	18.45 ↑	>100 ↑	19.70 ↑	>100 ↑	>100 ↑
TT (14–24 S)	13.47 ↓	14.28	15	14.32	14.41	15.54	14.97	14.72	12.63 ↓	13.22 ↓
D-Dimer (0–0.55 mg/L)	0.13	0.36	0.38	0.75 ↑	0.44	0.12	0.07	0.2	\	0.28
FDP (0–5 mg/L)	1.94	4	4.17	6.74 ↑	4.82	1.43	1.36	2.75	\	3.94
INR (0.8–1.2)	>8.16 ↑	>8.16 ↑	1.22 ↑	1.59 ↑	1.74 ↑	1.53 ↑	>8.16 ↑	1.63 ↑	>8.16 ↑	>8.16 ↑
FIB (2–4 g/L)	6.55 ↑	6.36 ↑	6.21 ↑	6.12 ↑	5.97 ↑	3.97	4.38 ↑	4.69 ↑	5.98 ↑	6.3 ↑
HGB (110–160 g/L)	114	114	98 ↓	97 ↓	96 ↓	107 ↓	108 ↓	\	\	40 ↓
PLT (100–300 × 10 ⁹ /L)	179	166	172	181	201	288	281	\	\	371 ↑
Neu% (50–70%)	76.7 ↑	73.3 ↑	76.7 ↑	66.3	72.5 ↑	68	73 ↑	\	\	93.3 ↑
BUN (1.7–8.3 mmol/L)	15.65 ↑	18.95 ↑	20.62 ↑	17.23 ↑	9.14 ↑	4.62	5.84	\	\	12.72 ↑
CRE (40–120 μmol/L)	437.3 ↑	522.4 ↑	494.7 ↑	214.6 ↑	134.1 ↑	92.5	94.9	\	\	103.5
BLD (-)	3+	\	\	\	3+	+-	+-	\	\	\
RBC in urine (0–10 /μL)	25,244 ↑	\	\	\	512.4 ↑	10.8	13.2 ↑	\	\	\
WBC in urine (0–15 /μL)	233 ↑	\	\	\	30.9 ↑	40.6 ↑	3.7	\	\	\

Note: The numbers in the parentheses represent the reference range for each parameter; “↑” or “↓” represent the value is higher or lower than the reference range, respectively; “>” represents the result has exceeded the quantitative upper limit; “\” represents that there is no record. (Abbreviations: APTT: activated partial thromboplastin time; PT: prothrombin time; TT: thrombin time; FDP: fibrinogen degradation product; INR: international normalized ratio; FIB: fibrinogen; HGB: hemoglobin concentration; PLT: platelet count; Neu%: neutrophil; BUN: blood urea nitrogen; CRE: creatinine; BLD: Urinary occult blood; RBC: red blood cell; WBC: white blood cell.)

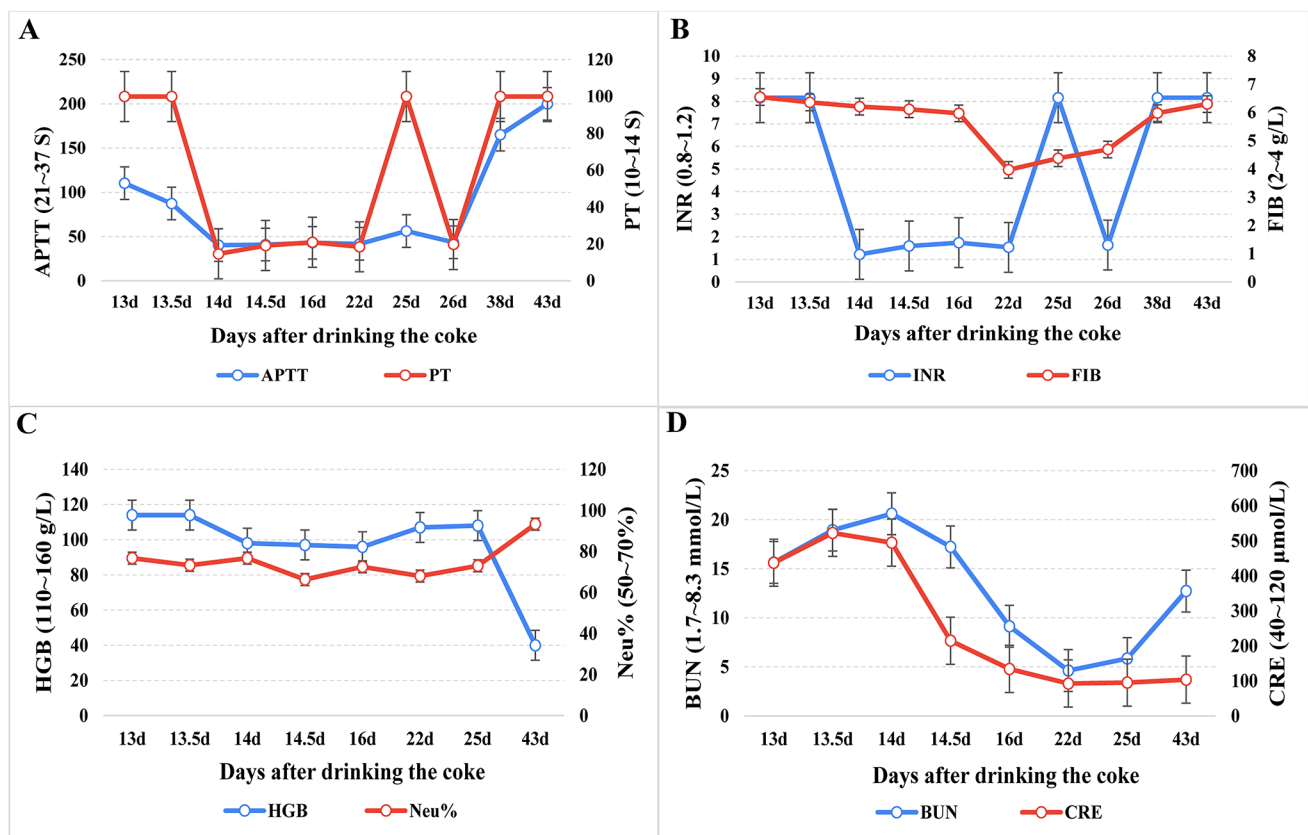


Fig. 2 The trend charts of some important clinical indicators over time. (A: trends of APTT and PT; B: trends of BUN and CRE; C: trends of HGB and Neu%; D: trends of INR and FIB.) (Abbreviations: APTT: activated partial thromboplastin time; PT: prothrombin time; INR:

international normalized ratio; FIB: fibrinogen; HGB: hemoglobin concentration; Neu%: neutrophil; BUN: blood urea nitrogen; CRE: creatinine.)

were combined and evaporated using a Termovap Sample Concentrator (MIULAB, China) at 60 °C. The residue was reconstituted with 100 µL of methanol (SCHARLAU, Spain) for further analysis.

LC conditions The LC system consisted of a LC-MS/MS 8045 (SHIMADZU, Japan). Samples were separated on an Agilent Eclipse Plus C18 column (100 mm×2.1 mm×3.5 µm, AGILENT, USA). Mobile phases were (A): 10 mmol/L ammonium acetate (ALADDIN, China) plus 5% methanol (SCHARLAU, Spain) and (B): 100% methanol (SCHARLAU, Spain) (All the above reagents were of MS grade), and delivered through the column (temperature + 25 °C) at a flow rate of 200 µL/min. The elution procedure lasted for 14 min, which started at 20% B for 1.5 min, raised to 80% B in 1.5 min and held for 2 min. Phase B then rose to 95% in 0.5 min and held for 3.5 min, returned to 20% in 1 min, then 20% B held for 4 min, to equilibrate the column before the next injection.

MS conditions MS data was acquired using electrospray ionization (ESI) source in negative ionization mode, utilizing multiple reaction monitoring (MRM) scan mode. The two pairs of precursor and product ions of the brodifacoum were 521.10 > 135.10 and 521.10 > 187.15 as shown in Fig. 3.

LC-MS/MS results The brodifacoum was successfully detected in the 43rd day blood sample with a quantitative result of 0.681 µg/mL.

Case investigation After conducting a thorough investigation, the police discovered that the deceased had borrowed money from the murderer and failed to repay it, resulting in animosity between the two persons. Additionally, due to the murderer's previous criminal activities of brodifacoum poisoning, he was aware that the human body does not succumb to poison immediately, and a prolonged death

following poisoning would assist him in evading suspicion. Therefore, the killer purchased three bottles of brodifacoum solution (as shown in Fig. S1) and poured all of them into a bottle of coke, subsequently administering it to the victim. According to the information on the package of brodifacoum solution, we can calculate that the effective ingredient of brodifacoum ingested by the patient was about 75 mg (5 g solution per bottle × 0.5% mg brodifacoum per mg of the solution × 3 bottles), which may serve as a reference for the potentially lethal dose of similar brodifacoum poisoning cases.

Discussion

Brodifacoum acts by impeding the synthesis of vitamin K. However, the vitamin K and clotting factors already present in the body could continue to function normally in the early period of poisoning. Hence, ingestion of brodifacoum poisoning does not immediately manifest symptoms; instead, there is typically a latent period during which symptoms may develop following brodifacoum poisoning and it was reported from 3 to 9 days [5]. After this latent period, patients commonly experience bleeding in various locations, such as vaginal and rectal bleeding [6], gastric and pulmonary hemorrhage [7], hematuria, hematemesis, melena, and other sites of bleeding [8]. Among these manifestations, hematuria has been reported as a significant clinical symptom [5]. In this case, hematuria appeared 10 days after the individual consumed the brodifacoum-laced coke and persisted for 3 days until relevant treatment was administered. Moreover, the patient's physical condition improved after continuous treatment for 2 weeks. However, once the treatment was discontinued, the condition deteriorated again, highlighting the long half-life of brodifacoum which was reported at the range from 20 to 130 days [9].

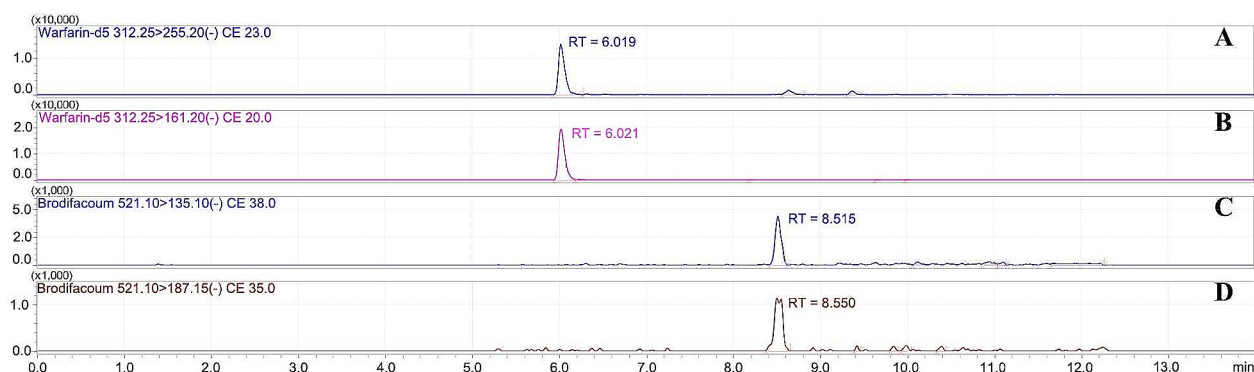


Fig. 3 The chromatogram of brodifacoum in the victim's blood (A: Warfarin-d5 as internal standard, 312.25 > 255.20; B: Warfarin-d5 as internal standard, 312.25 > 161.20; C: Brodifacoum, 521.10 > 135.10; D: Brodifacoum, 521.10 > 187.15). (Abbreviations: RT: retention time)

In cases where there is a suspicious history, bleeding in multiple parts of the body, prolonged PT and APTT, and a high INR, brodifacoum poisoning should be considered as a diagnosis [10]. In this case, based on the patient's clinical manifestations and blood clotting function tests, the diagnosis of brodifacoum poisoning was inferred suspiciously, which was finally confirmed with the assistance of the LC-MS/MS results. The treatment for poisoned patients typically involves the administration of vitamin K₁ (via intravenous injection or oral administration) and fresh frozen plasma [11–13]. It is important to note that the initial dosage of vitamin K₁ should be adjusted based on the clinical manifestations, with reported daily doses ranging from 15 to 300 mg [13, 14]. Additionally, the therapy should be continued until all symptoms disappear and the brodifacoum levels in the blood fall below 0.010 µg/mL [7, 15]. In this case, although the patient received timely and effective clinical diagnosis and treatment of administration of vitamin K₁ and fresh frozen plasma during both hospitalizations, tragic consequences occurred due to the discontinuation of therapy. It is crucial to ensure that the treatment is sustained to achieve complete recovery and prevent relapse.

Timely performance of mass spectrometry is essential to identify the cause accurately and develop a more precise therapeutic approach. However, many laboratories do not possess the capability to analyze brodifacoum qualitatively and quantitatively. In this case, a quantitative result of 0.681 µg/mL was successfully detected by a mass spectrometric method in the patient's blood sample which was collected for the blood routine and biochemical tests on the 43rd after the individual consumed the brodifacoum-laced coke. Comparing this concentration with other reported concentrations of brodifacoum in blood, for example, 1.302 µg/mL on the 5th day [7], 0.170 µg/mL on the 30th day [15], 0.442 µg/mL on the 20th day [10], and 0.032 µg/mL on the 9th day [16], the concentration in this case was relatively high. Besides, there were published studies reporting fatal poisonings in patients with plasma brodifacoum levels ranging from 0.030 to 0.786 µg/mL [17]. It had also been observed that coagulopathies can occur with ingested doses of brodifacoum ranging from 1 to 75 mg and can persist for up to 18 months after a single ingestion [18]. In this case, based on the information on the packaging box, the estimated effective brodifacoum content was approximately 75 mg. Therefore, we believe that the dose of brodifacoum ingested had exceeded the dosage threshold for developing coagulopathy. Furthermore, due to the lack of subsequent medical treatment, the coagulopathy worsened, ultimately leading to death.

In forensic practice, determining the cause of death often involves conducting a thorough and systematic autopsy. However, in cases where autopsies are not performed, the

approach to determining the cause of death relies on the patient's clinical history and the results of blood toxicology tests. So, we would like to report this case and highlight the clinical history and blood toxicology results which may contribute to the existing knowledge and understanding of brodifacoum poisoning.

Conclusions

This paper presents a rare case of death resulting from brodifacoum poisoning. Despite the absence of a comprehensive autopsy, the cause of death was eventually established through careful analysis of the patient's detailed case data, examination reports, and blood toxicology analysis result. The aim of this paper is to provide valuable references for the diagnosis, treatment, and determination of cause in similar brodifacoum poisoning cases.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00414-024-03198-z>.

Declarations

Ethical approval This study does not need ethical approval.

Informed consent Not required.

Conflict of interest The authors declare no competing interests.

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